

where  $x_i$  are real coefficients that can be positive or negative such that Eq. (1) is a quasiprobability representation.

It follows that the expectation value of an observable  $O$  for the target operation can be written in terms of the expectation value of the noisy operations as

$$\begin{aligned} \text{Tr}[OU^t(\rho)] &= \text{Tr}\left[O \sum_i x_i \mathcal{U}_i^n(\rho)\right] \\ &= \sum_i x_i \text{Tr}[O\mathcal{U}_i^n(\rho)]. \end{aligned} \quad (2)$$

The expectation values  $\text{Tr}[O\mathcal{U}_i^n(\rho)]$  associated with noisy channels  $\mathcal{U}_i^n$  can be estimated using Monte Carlo sampling. Each noisy operation  $\mathcal{U}_i^n$ , specified by index  $i$ , is applied with probability  $|x_i|/\sum_i |x_i|$ , and the resulting expectation value is multiplied by  $\text{sign}(x_i) \sum_i |x_i|$  [26, 31]. From this, an estimate of the ideal expectation value can be calculated up to an accuracy  $\epsilon$  and probability greater than  $1 - \delta$ , where

$$\delta = 2 \exp\left(\frac{-N\epsilon^2}{2(\sum_i |x_i|)^2}\right) \quad (3)$$

and the number of samples  $N$  is determined by Hoeffding's inequality to be

$$N = \frac{2}{\epsilon^2} \left(\sum_i |x_i|\right)^2 \ln\left(\frac{2}{\delta}\right). \quad (4)$$

Thus, the number of samples scale as  $\mathcal{O}(\lambda^2/\epsilon^2)$  up to logarithmic factors, with  $\lambda = \sum_i |x_i|$ . The quantity  $\lambda^2$  is referred to as the *sampling overhead* of using this method over finding the expectation value of the ideal target channel directly [26, 27].

### III. METHODOLOGY

#### A. Linear Combination of Channels (LCC) Decomposition

We first demonstrate how the quasiprobability method can be used to decompose target  $\mathcal{Z}$ -rotation channels. In the simplest case, we consider a diagonal Cliffords +  $\mathcal{T}$  channel decomposition, forming an overcomplete basis set to ensure that we can optimise this decomposition with respect to the sampling overhead.

Consider the action of a small-angle  $\mathcal{Z}$ -rotation channel  $\mathcal{R}_z^\theta(\rho) = R_z(\theta)\rho(R_z(\theta))^\dagger$  acting on a state represented by density matrix  $\rho$  in Fig. 1, where  $R_z(\theta) = \exp(-i(\theta/2)Z)$  is a gate representing a rotation about the  $Z$ -axis with angle  $\theta$ . In this geometric picture, we label eight channels corresponding to the action of a basis set of gates

$$\begin{aligned} G &= \{T^k : 1 \leq k \leq 8\} \\ &= \{I, T, S, ST, Z, ZT, ZS, ZST\}, \end{aligned} \quad (5)$$

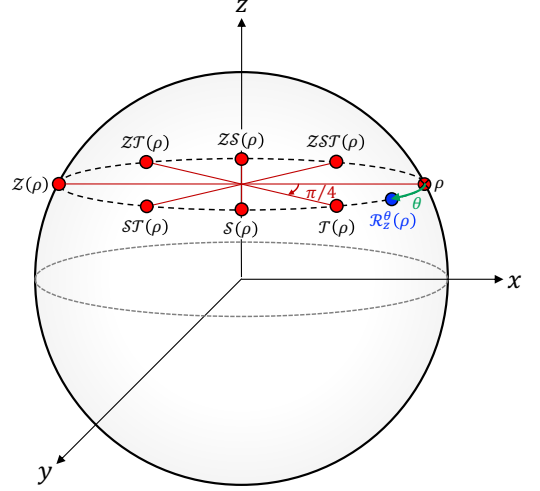


FIG. 1. Action of  $\mathcal{T}^k$  rotation channels (where  $1 \leq k \leq 8$ ) on a density matrix  $\rho$ . The eight operations (in red) form the initial basis set  $\mathcal{G}$ . The target rotation channel  $\mathcal{R}_z^\theta(\rho)$  is shown in blue.

which forms a cyclic group  $(G, \times)$  generated by the  $T$  gate under multiplication. We represent these gates in their channel representation to form a set  $\mathcal{G}$  such that

$$\mathcal{G} = \{\mathcal{T}^k(\rho) = (T^k)\rho(T^k)^\dagger : 1 \leq k \leq 8\}. \quad (6)$$

The LCC decomposition is given by

$$\mathcal{R}_z^\theta(\cdot) = \sum_{U \in \mathcal{G}} x_U U(\cdot) U^\dagger = \sum_{U \in \mathcal{G}} x_U \mathcal{U}(\cdot), \quad (7)$$

where  $\mathcal{R}_z^\theta(\cdot)$  represents the single qubit  $\mathcal{Z}$ -rotation channel for the  $R_z(\theta)$  gate. This is a quasiprobability decomposition such that the coefficients  $x_U$  are real and satisfy  $\sum_{U \in \mathcal{G}} x_U = 1$ . The quantity  $\lambda$  is defined as the  $l_1$ -norm of the coefficients in this decomposition, that is

$$\lambda = \|\mathbf{x}\|_1 = \sum_{U \in \mathcal{G}} |x_U|, \quad (8)$$

such that  $\lambda^2$  is the sampling overhead of each decomposition from Hoeffding's inequality [25, 27].

It can be noted that the decomposition in Eq. (7) is not unique over a chosen basis set and so it is important to minimise the sampling overhead over different solutions. We define the optimal decomposition for a given  $\mathcal{Z}$ -rotation channel as the decomposition into elements of  $\mathcal{G}$  with the minimum  $\lambda$ , which we denote as  $\Lambda_{\mathcal{G}}(\mathcal{R}_z^\theta)$ . This can be written as

$$\Lambda_{\mathcal{G}}(\mathcal{R}_z^\theta(\cdot)) = \min \left\{ \lambda = \|\mathbf{x}_U\|_1 \left| \mathcal{R}_z^\theta(\cdot) = \sum_{U \in \mathcal{G}} x_U \mathcal{U}(\cdot) \right. \right\}. \quad (9)$$

By definition,  $\Lambda_{\mathcal{G}}(\mathcal{R}_z^\theta) \geq 1$  where  $\Lambda_{\mathcal{G}}(\mathcal{R}_z^\theta) = 1$  if  $\mathcal{R}_z^\theta \in \mathcal{G}$  for some group  $\mathcal{G}$  representing the basis set of channels. The solution to Eq. (9) is equivalent to minimising the